Option Pricing in Python:
Six Models Checked Against Each Other

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1 The Core Idea

A pricing model is only useful if its output can be trusted. The repository builds six small pricers from scratch (Black-Scholes, binomial tree, Greeks, a barrier option, delta hedging, option strategies) and checks each one against a second method.

All notebooks use the same inputs: spot $S = 100$, strike $K = 100$, rate $r = 3\%$ and volatility $\sigma = 20\%$. Maturity is one year, except for the hedging and strategy notebooks (three months).

- **Closed form against numerics:** the tree converges to Black-Scholes, the Greeks match finite differences, the barrier formula matches Monte Carlo.
- **Model against hedge:** a short call hedged with the model's own delta should end with a P&L close to zero, and the error should shrink with more rebalancing.

2 Why It Matters

- **A model price is a claim that can be tested.** A risk manager who uses a pricer needs to know how far it is from a second method. Here the 1,000-step tree is 0.0020 below Black-Scholes on a price of 9.4134, an error of two parts in ten thousand.
- **The contract changes the price.** The American put is worth 6.74 against 6.45 for the European one, and a down-and-out call with a barrier at 90 costs 7.74 against 9.41 for the vanilla call, 18% less. A trader who prices an exotic as a vanilla is off by that much.
- **Hedging removes direction, not risk.** A call sold and delta hedged four times over three months ends with a standard deviation of 1.57 on a premium of 4.36. Hedging daily (63 times) brings it to 0.435, and 252 times to 0.22. What is left is the cost of discrete trading.
- **Strategies are positions on volatility.** The long straddle gains 0.396 per vol point and the butterfly loses 0.155. A strategy chosen as a bet on direction carries a volatility bet that the trader may not intend.

3 Assumptions

Each line states what the code assumes and which way the result moves if it is wrong.

- **Black-Scholes world.** The price is geometric Brownian motion with constant volatility of 20% and a constant continuous rate of 3%, no dividends, no transaction costs and no bid-ask spread. Real prices have a smile, jumps and costs, so a model price is a reference level and not a quote.
- **Implied volatility needs a bracket.** Brent's method searches $[10^{-4}, 5]$. A call below its lower bound $S - Ke^{-rT} = 2.9554$ has no solution and the code raises an error. Above the bound the inversion is smooth: 12.00 gives 26.68%, and each 0.5 of price moves the volatility by about 1.3 points.
- **Tree.** Cox-Ross-Rubinstein with $u = 1/d = e^{\sigma\sqrt{\Delta t}}$ and the risk-neutral probability. It prices with no dividends, so an American call equals the European call. The American value is the maximum of continuation and intrinsic value at every node.
- **Greek units and finite differences.** Vega is per vol point, theta per calendar day (divided by 365) and rho per 1% of rate. The numerical Greeks use a bump of 10^{-4}. The check in the notebook asserts agreement to 10^{-3}, and the actual gap is below 10^{-6}.
- **Barrier.** The formula is for a continuously watched barrier below the strike, with no rebate and no dividends. The Monte Carlo watches the barrier 252 times, with 100,000 paths and seed 1. A less frequent monitoring means a higher price.
- **Hedging experiment.** The hedge uses the true volatility, so the model is correct by construction. Paths are simulated with the risk-neutral drift r, and a real-world drift changes little (Section 7). 2,000 paths per case, one seed (7), no costs. With a wrong volatility the result changes a lot (Section 7).
- **Strategies.** Profit is computed at expiry only, from Black-Scholes premiums at 20% volatility, compounded at r, on a grid of 1,601 prices between 60 and 140. A strategy marked "unlimited" is still rising at 140. Break-evens have the precision of the grid (0.05).

4 Method

4.1 Black-Scholes and implied volatility

With $d_1 = \dfrac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$ and $d_2 = d_1 - \sigma\sqrt{T}$:

$$C = S\,N(d_1) - Ke^{-rT}N(d_2), \qquad P = Ke^{-rT}N(-d_2) - S\,N(-d_1).$$

The call is the present value of owning the stock when the option ends in the money, minus the strike paid in that case. Put-call parity $C - P = S - Ke^{-rT}$ holds for any volatility and is used as a check. The implied volatility is the root of $C(\sigma) - C_{mkt} = 0$ found with Brent's method on [0.0001, 5]. The price is increasing in σ, so the root is unique.

4.2 Binomial tree (Cox-Ross-Rubinstein)

With n steps and $\Delta t = T/n$:

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = 1/u, \quad p = \frac{e^{r\Delta t} - d}{u - d}, \quad V_t = e^{-r\Delta t}\left(pV_{t+1}^{up} + (1-p)V_{t+1}^{down}\right).$$

For an American option, the value at each node is $\max(V_t,$ intrinsic value). The tree matches the first two moments of the log return, so it tends to Black-Scholes as n grows. Because the

strike sits on or off the grid of terminal prices depending on whether n is even or odd, the error alternates in sign.

4.3 Greeks

Closed forms for the call, with φ the normal density, in the units used by the code:

$$\Delta = N(d_1), \qquad \Gamma = \frac{\varphi(d_1)}{S\sigma\sqrt{T}}, \qquad \mathcal{V} = \frac{S\varphi(d_1)\sqrt{T}}{100},$$
$$\Theta = \frac{1}{365}\left(-\frac{S\varphi(d_1)\sigma}{2\sqrt{T}} - rKe^{-rT}N(d_2)\right), \qquad \rho = \frac{KTe^{-rT}N(d_2)}{100}.$$

Vega is per vol point, theta per calendar day and rho per 1% of rate. Each is compared with a central finite difference with a bump of 10^{-4}. Gamma and theta have the same origin: a long option pays theta every day to own gamma.

4.4 Down-and-out call

For a barrier $H \le K$ watched continuously, the price is the vanilla call minus the down-and-in call. With $\lambda = (r + \frac{1}{2}\sigma^2)/\sigma^2$ and $y = \dfrac{\ln(H^2/(SK))}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T}$:

$$C_{DI} = S\left(\tfrac{H}{S}\right)^{2\lambda} N(y) - Ke^{-rT}\left(\tfrac{H}{S}\right)^{2\lambda-2} N(y - \sigma\sqrt{T}), \qquad C_{DO} = C - C_{DI}.$$

The Monte Carlo uses 100,000 paths of 252 daily steps (seed 1) and sets the payoff to zero if any daily price is at or below H. Daily monitoring is less strict than continuous monitoring, so the simulated price is higher. The Broadie-Glasserman-Kou correction moves the barrier to $H\,e^{-0.5826\,\sigma\sqrt{\Delta t}}$ in the formula.

4.5 Delta hedging and strategies

The seller of a call receives the premium, buys Δ shares and rebalances at n equal dates. Cash grows at r. At expiry the final P&L is

$$\text{P\&L} = \text{cash}_T + \Delta_{n-1}S_T - \max(S_T - K, 0).$$

There are no costs or dividends. The test uses a 3-month at-the-money call, 2,000 simulated paths per case and seed 7. The error is the part of the option's randomness that a delta position cannot track between two dates: it grows with gamma and with the squared move.

For strategies, each leg is a (type, strike, quantity). The profit at expiry is $\sum_i q_i\,\text{payoff}_i(S_T) - \text{cost} \times e^{rT}$, where the cost is the sum of Black-Scholes prices. Break-evens are the sign changes on a grid of 1,601 prices between 60 and 140.

5 Method Comparison

Method	Strength	Weakness
Closed form	Exact and instant, gives Greeks	Only simple payoffs, constant volatility
Binomial tree	Handles early exercise	Error $\approx 2/n$; sign alternates with n
Monte Carlo	Any path-dependent payoff	Error $\propto 1/\sqrt{N}$; discrete monitoring bias
Finite differences	Checks any formula	Noise when the bump is too small

Use the closed form whenever it exists and keep the tree or the simulation as an independent check. The tree is the only method here that prices the American put. Monte Carlo is the choice when the payoff depends on the path, as for the barrier, and the correction of the monitoring bias is worth the extra line of code.

6 Results

Tree against Black-Scholes (ATM call, 1 year, Black-Scholes price 9.4134):

Steps	Tree price	Error
10	9.2179	−0.1956
100	9.3936	−0.0198
1000	9.4114	−0.0020

The error falls in proportion to $1/n$ (ten times more steps, ten times smaller error), with $n \times$error close to -1.98 for even n. The three values above are all even. With odd n the sign flips: at $n = 1001$ the error is $+0.0019$, and the average of the trees at 1,000 and 1,001 steps is about 0.00005 from Black-Scholes. A tree reported at one n carries a bias of this size, so the notebook's check (< 0.01 at 1,000 steps) is loose.

American against European puts (500 steps). An American call without dividends has the same value as the European call (9.4094 for both).

Strike	European put	American put	Early exercise premium
80	0.8607	0.8830	0.0223
100	6.4540	6.7411	0.2871
120	19.2195	20.6068	1.3873
140	36.4932	40.0000	3.5068

At $K = 140$ the American put is worth its intrinsic value of 40: it is optimal to exercise at once. The premium comes from the interest earned on the strike received early. It is zero if the rate is zero, 0.076 at 1%, 0.287 at 3% and 0.636 at 6% (ATM put, 1 year). It hardly depends on volatility (0.301, 0.287 and 0.286 at 10, 20 and 40%), but it grows with maturity (0.058 at 3 months, 0.287 at 1 year, 0.640 at 2 years). The table does not include dividends or a rate smile, and both would change the early exercise boundary.

Greeks. The formula and the finite difference agree to five decimals for the ATM call: delta 0.59871, gamma 0.01933, vega 0.38667, theta −0.01474 and rho 0.50457. The largest gap is

4.6×10^{-7} on gamma, the second derivative, which is the most sensitive to the bump. Maturity changes the picture: with the same inputs at $T = 0.1$, gamma is 0.06288 against 0.01933 at one year, and theta is -0.0386 a day against -0.0147. A short-dated option has more gamma and more theta, and the two are the price of each other.

Barrier ($H = 90$): the vanilla call is 9.4134. The down-and-out call is 7.7402 by formula. Monte Carlo gives 8.0034 (95% interval ± 0.0869). The formula with the adjusted barrier gives 7.9700, which is inside the Monte Carlo interval.

Monitoring dates	Monte Carlo	Adjusted formula	Adjusted barrier
12	8.5527	8.5900	87.02
52	8.2633	8.2109	88.56
252	8.0034	7.9700	89.34
504	7.9401	7.9056	89.53

The price falls towards the continuous value of 7.7402 as the monitoring becomes more frequent. The corrected formula follows the simulation at every frequency, and the gap (between -0.04 and $+0.05$) stays inside the interval of ± 0.087. The barrier is far from remote: 55.5% of the simulated daily paths touch 90. Of the knocked-out paths, 76% finish below the strike, where the vanilla call pays nothing (seed 1, 100,000 daily paths). The barrier therefore removes 15% of the vanilla value in the simulation, and 18% in the continuous formula, not 55%. The effect of the barrier depends on the volatility: the down-and-out call is worth 99% of the vanilla call at 10% volatility, 82% at 20% and 53% at 40%.

Delta hedging (call sold for 4.3576):

Rebalances	Mean P&L	Std P&L	Formula sd
4	+0.0445	1.5724	1.754
13	+0.0112	0.9449	0.973
63	−0.0004	0.4349	0.442
252	+0.0023	0.2199	0.221

The last column is the Derman–Kamal leading-order formula $\sqrt{\pi/4}\,\mathcal{V}\sigma/\sqrt{N}$ with call vega $\mathcal{V} = 19.79$ per 1.00 of volatility, added for comparison (it is not in the code). The formula is 12% above the simulation at 4 rebalances and 0.5% above at 252. The mean is zero within sampling error: the standard error is the standard deviation over $\sqrt{2000}$, 0.035 for 4 rebalances and 0.005 for 252, so 0.0445 is not different from zero. The error has a fatter left tail than right: for 4 rebalances the 1st percentile is -4.49 and the 99th is $+3.17$, with a worst path at -7.23. The standard deviation moves by about 1% across seeds at 63 rebalances (0.430, 0.432, 0.438 for three other seeds, against 0.435).

Strategies (3 months, strikes in brackets; a negative cost is a credit; vega is the change in the strategy value for one vol point, computed with a bump of 0.1 vol point):

Strategy	Cost	Max profit	Max loss	Break-even	Vega
Bull call spread (95/105)	5.07	4.89	−5.11	100.1	−0.024
Long straddle (100)	7.97	unlimited	−8.03	92.0, 108.0	+0.396
Long strangle (95/105)	3.98	unlimited	−4.01	91.0, 109.0	+0.350
Butterfly (90/100/110)	3.66	6.31	−3.69	93.7, 106.3	−0.155
Iron condor (85/90/110/115)	−1.07	1.08	−3.92	88.9, 111.1	−0.109

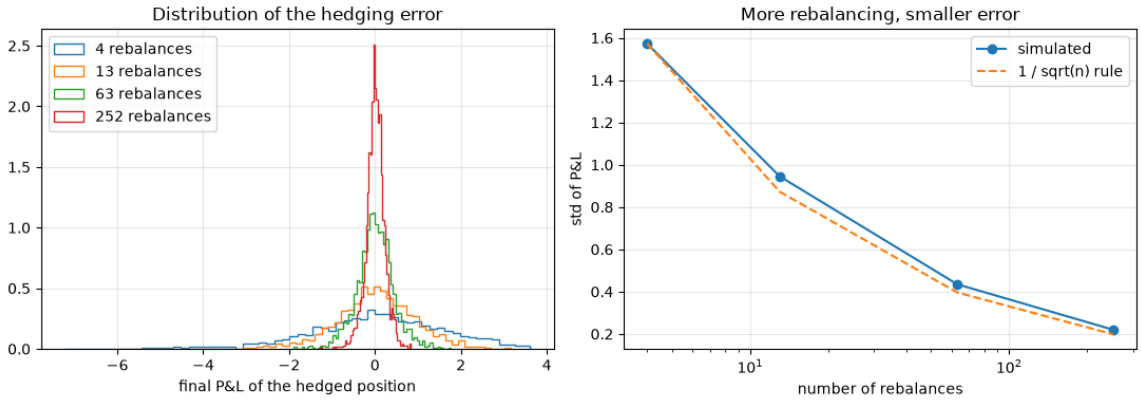


Figure 1: Distribution of the hedging error (left) and its standard deviation against the $1/\sqrt{n}$ rule (right).

The maximum loss of a bought strategy is the premium compounded at r (7.97 gives 8.03 for the straddle), and the maximum profit of the butterfly is the wing width minus the compounded cost ($10 - 3.69 = 6.31$). The sign of the vega says what each structure bets on. The straddle and strangle gain from higher volatility, and the butterfly and condor gain from lower volatility and a quiet market. The bull call spread is almost neutral to volatility (-0.024), which makes it a cleaner view on direction. Break-evens are the exact roots of the profit at expiry, rounded to one decimal (the notebook finds them on a grid of 0.05 and prints the same values). The table gives them at expiry, so it does not say how the strategy behaves before expiry or how likely each outcome is.

7 Worked Example

Black-Scholes by hand for $S = K = 100$, $T = 1$, $r = 3\%$, $\sigma = 20\%$:

$$d_1 = \frac{0 + (0.03 + 0.02) \times 1}{0.20} = 0.25, \qquad d_2 = 0.25 - 0.20 = 0.05,$$
$$N(d_1) = 0.598706, \qquad N(d_2) = 0.519939, \qquad Ke^{-rT} = 97.0446,$$
$$C = 100 \times 0.598706 - 97.0446 \times 0.519939 = 59.8706 - 50.4572 = 9.4134.$$

The put is $97.0446 \times 0.480061 - 100 \times 0.401294 = 46.5873 - 40.1294 = 6.4579$ (6.4580 with unrounded inputs). Parity: $9.4134 - 6.4580 = 2.9554 = 100 - 97.0446$. The numbers $N(d_1)$ and $N(d_2)$ also give the delta and, through $KTe^{-rT}N(d_2)/100$, the rho of 0.50457 in the Greeks results above.

The Greeks satisfy the Black-Scholes equation. For the same call, theta per year is $-0.01474 \times 365 = -5.3804$, so

$$\Theta + \tfrac{1}{2}\sigma^2 S^2 \Gamma + rS\Delta - rC = -5.3804 + 3.8667 + 1.7961 - 0.2824 = 0.0000.$$

The check ties the Greeks together: the theta paid is the gamma earned plus the carry on the delta position.

Barrier correction. For daily monitoring, $\sigma\sqrt{\Delta t} = 0.20 \times \sqrt{1/252} = 0.01260$, so

$$H_{adj} = 90 \times e^{-0.5826 \times 0.01260} = 90 \times e^{-0.00734} = 89.34.$$

Lowering the barrier makes the option less likely to be knocked out, so the price rises from 7.7402 to 7.9700.

The $1/\sqrt{n}$ rule. From 4 rebalances (std 1.5724), the rule predicts $1.5724 \times \sqrt{4/252} = 0.198$ at 252 rebalances. The simulation gives 0.2199, about 11% higher.

Sensitivity. All lines rerun the notebook code.

- *Implied volatility.* For call prices of 9.4134, 10, 12, 15 and 20, the implied volatility is 20.0, 21.5, 26.7, 34.4 and 47.5%. The volatility moves by about 2.6 points per unit of price between 12 and 20. A price below 2.9554 has no implied volatility.
- *Volatility level.* At $\sigma = 10$, 20, 30, 40% the ATM call is 5.58, 9.41, 13.28 and 17.14. The step is 3.8 to 3.9 for each 10 points, close to the vega of 0.39 per point, and the 0.4 $S\sigma\sqrt{T}$ rule of thumb (8.0 at 20%) works at a zero rate (7.97).
- *Strategy costs.* At $\sigma = 15$, 20, 30% the straddle costs 5.99, 7.97 and 11.93, the iron condor pays a credit of 0.52, 1.07 and 2.01 and the butterfly costs 4.61, 3.66 and 2.55.
- *Hedging with the wrong volatility.* The call is sold and hedged at 20% with 63 rebalances, while the paths have a volatility of 15, 20, 25 and 30%. The mean P&L is +0.998, +0.021, −0.996 and −1.960, against price gaps $C(20\%) - C(\sigma_{real})$ of +0.989, 0, −0.990 and −1.980. The seller who underprices volatility by 5 points loses about a fifth of the premium (4.36), whatever the hedge. The standard deviation is 0.46, 0.44, 0.70 and 1.08. A real-world drift of −15%, 3% or +15% leaves the mean at −0.029, +0.003 and −0.001 (standard error 0.01): the delta hedge removes the drift.

8 Limits and Extensions

- **Textbook assumptions:** constant volatility and rate, no dividends, no transaction costs, and continuous trading in the limit. Real markets have a smile and costs.
- **Small samples:** the hedging test uses 2,000 paths per case and one seed. The mean P&L values of 0.0445 against 0.0023 are noise of the order of the standard error (about 0.035 for 4 rebalances).
- **Barrier monitoring:** the Monte Carlo with 252 steps sits 0.033 above the corrected formula, inside its interval. The correction is an approximation.
- **No market data:** all prices are model prices, so the notebooks check internal consistency, not agreement with traded prices. These are model outputs, not recommendations.

Three extensions would address these limits. A dividend yield and a local or stochastic volatility would make the prices comparable with traded quotes. Using antithetic or control variates, with the vanilla call as control, would cut the Monte Carlo interval of the barrier price at no extra cost. A hedging test with transaction costs would show the frequency that balances the error against the trading bill.

9 Conclusion

Each pricer agrees with its check to the precision that the check allows, which is the case for a model with no market data: the useful lesson is the size of the gaps, for instance a tree bias of 0.002, a monitoring bias of 0.26 on the barrier and a hedging error of 0.435 after daily hedging (0.22 with 252 rebalances). A risk manager should ask for these gaps next to every model price.